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Modelling the effect of Docosa Hexaenoic Acid (DHA) in cancer therapy

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ABSTRACT

Background: Docosahexaenoic Acid (DHA), an omega-3 fatty acid, has demonstrated potential anti-cancer properties, including anti-proliferative and pro-apoptotic effects on tumor cells. Understanding the impact of DHA on tumor growth dynamics can inform therapeutic strategies and optimize treatment efficacy. Tumor growth models, including the exponential, logistic, and Gompertz models, provide frameworks for quantifying these effects.

Objective: This study aims to integrate the effects of DHA into various tumor growth models to evaluate its impact on tumor dynamics and to provide insights into its potential as a therapeutic agent in cancer treatment.

Methods: The effects of DHA were incorporated into the exponential, logistic, and Gompertz growth models by modifying the intrinsic growth rate parameter. Experimental data from in vitro and in vivo studies were used to estimate the reduction in growth rates due to DHA treatment. The modified models were analyzed to determine the extent of DHA's impact on tumor growth.

Results: Incorporation of DHA into the exponential growth model indicated a significant reduction in the intrinsic growth rate (r), reflecting its inhibitory effect on tumor proliferation. The logistic growth model, which accounts for environmental carrying capacity, showed that DHA not only reduced the growth rate but also potentially affected the carrying capacity (K), suggesting an influence on the tumor microenvironment. The Gompertz model, suitable for long-term growth dynamics, demonstrated that DHA decreased the initial growth rate and might affect the deceleration parameter (bb), indicating sustained inhibition of tumor growth over time.

Conclusion: Integrating DHA into tumor growth models provides a quantitative understanding of its anti-tumor effects. DHA effectively reduces tumor growth rates across different models, highlighting its potential as a valuable component of cancer therapy. Further research should focus on optimizing DHA dosage and administration schedules, exploring its synergistic effects with other treatments, and validating these findings in clinical settings.

INTRODUCTION

Tumor growth models are mathematical and computational frameworks used to describe and predict the dynamics of tumor development and progression. These models are critical for understanding cancer biology, planning treatment strategies, and designing clinical trials [1, 2]. They range from simple models describing overall tumor growth to complex models incorporating various biological processes and interactions [3, 4].

Tumor growth models are mathematical representations that describe the dynamics of tumor development and progression [5, 6]. These models are essential in understanding tumor biology, predicting treatment outcomes, and optimizing therapeutic strategies [7-13].

In this paper the authors are presenting the mathematical models for tumor growth and the incorporation of Docosa Hexaenoic Acid (DHA) in Tumor Growth Models:

Types of tumor growth models

There are numerous tumor growth models, each with its own strengths and limitations.